

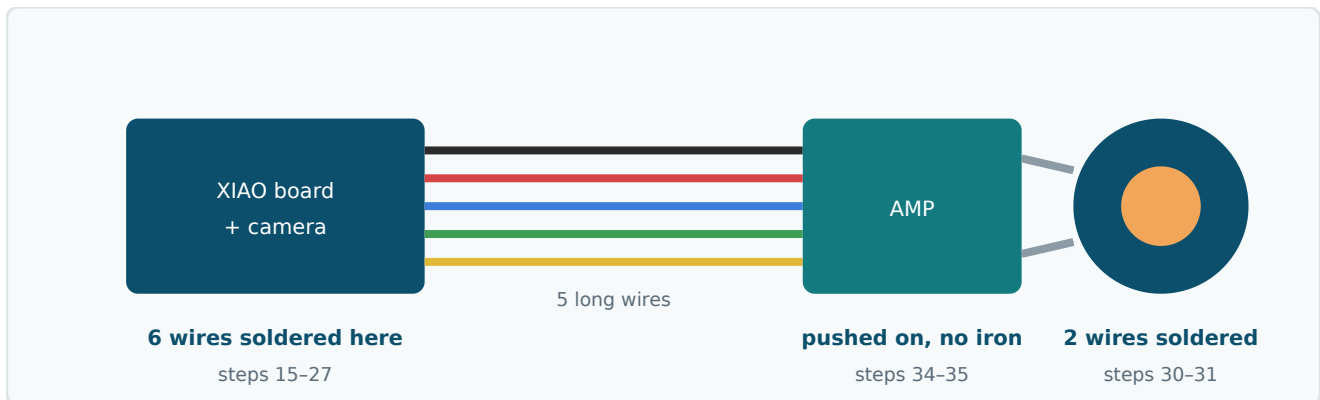
Every Single Step

42 steps. One action each. Nothing skipped.

How to use this. Do step 1. Read the green "you should see" line and check it actually happened. Only then do step 2. If the green line didn't happen, the red line tells you what to do. **Never skip ahead** — every step assumes the one before it worked.

Two safety things, once. The iron is 350°C and looks identical hot or cold — **assume it is always hot**, and put it back in its stand every single time you let go of it. Solder fumes are unpleasant: work somewhere with a window open, and don't lean directly over the joint.

WHAT YOU'RE MAKING



Total: 9 soldered joints. Six on the board, two on the speaker, one on a coin. Everything else is cutting, stripping, plugging in, and checking. Most of these 42 steps involve no iron at all.

PART 1 — Set up your bench

No iron yet. Steps 1–8.

1 Unplug the USB cable from the XIAO board. ☐

If it isn't plugged in, good — leave it that way.

You should see: nothing connected to the board at all.

Why: soldering a powered board can destroy it.

2 Put the board on the table with the USB socket pointing away from you. ☐

The silver USB socket is at one short end. Point that end away.

You should see: a row of 7 small silver pads down the left edge, and 7 down the right.

3 Tape the board down. ☐

A loop of tape under it, or tape over each end. Any tape.

You should see: the board doesn't slide when you nudge it.

Why: one hand holds the iron, one holds the solder. You have no hand left for the board.

4 Find the writing next to the pads. ☐

Tiny text: `D0 D1 D2 D3 D4 D5 D6` on one edge, and `5V GND 3V3 D10 D9 D8 D7` on the other. Use your phone camera zoomed in if you can't read it.

You should see: you can point at the pad labelled `3V3` and the one labelled `D8`.

If you can't find these: stop and message me a photo. Everything else depends on this.

5 Plug the soldering iron in and set it to 350°C. ☐

No temperature dial? Just plug it in. It'll settle wherever it settles and that's usually fine.

You should see: it takes 2–5 minutes to heat. Wait.

6 Wet the sponge in the iron's stand. ☐

A splash of water. Damp, not soaking — squeeze the excess out.

You should see: a damp sponge that doesn't drip.

7 Once hot, wipe the iron's tip on the wet sponge. ☐

One quick drag. It will hiss and steam — that's correct.

You should see: the very tip goes shiny silver instead of dull black.

8 Touch solder to the tip until a bead forms, then wipe again. ☐

This is called "tinning the tip". Melt a bit of solder onto it, wipe most off.

You should see: the tip stays bright silver and shiny.

If it stays black: the tip is oxidised. Repeat 7–8 twice more. A black tip will not transfer heat and nothing else will work.

PART 2 — Prepare the wires

Still no iron. Steps 9–14.

9 Put your glasses on the table and measure along one arm. ☐

From the hinge (where the arm meets the lens) to the point that sits behind your ear. Use a ruler.

You should see: a number, roughly 10–14 cm. **Write it down.**

10 Add 2 cm to that number. This is your cut length. ☐

So 12 cm measured → cut at 14 cm.

You should see: one number written down that you'll use five times.

Why: wires always need more slack than you think. Too long tucks away; too short is unfixable.

11 Cut FIVE wires to that length: one red, one black, one yellow, one green, one blue. ☐

If your jumper wires have plastic plugs on the ends, cut those off — you need bare wire.

You should see: five wires of equal length in five different colours.

No blue or green? Any five distinct colours work. Just write down which colour you chose for which job and stick to it.

12 Cut ONE more wire, white, about 5 cm long. ☐

Much shorter — this one never leaves the front pod.

You should see: six wires total. Five long, one short.

13 Strip 3 mm of plastic off BOTH ends of ALL SIX wires. ☐

3 mm is about the width of a matchstick. Strippers if you have them; otherwise score gently with a knife and pull the plastic off.

You should see: 12 bare metal ends, each about 3 mm long.

If you cut through the metal: snip that end off and strip again. It's fine, you have slack.

14 Twist each bare end tightly between your fingers. ☐

The metal is many thin strands. Roll them together so they form one solid point.

You should see: 12 neat twisted points, no stray strands sticking out sideways.

Why: one loose strand touching the next pad is a short circuit.

You're a third of the way through and haven't touched the iron. That's normal.

Preparation is most of the job; the soldering itself is fast.

PART 3 — Put solder on the six pads

Iron time. Steps 15-20.

The move, for all six. Touch the iron tip to the pad. **Count "one, two"**. Now touch solder to the *pad* (not the iron). It melts and spreads. Take the solder away, then the iron. Total: about 3 seconds.

- 15

Put solder on the pad labelled `GND` .

☐

Right-hand edge, second one down from the USB end.

You should see: a small shiny silver dome covering the pad.

If the solder balls up and rolls off: the pad wasn't hot enough. Hold the iron on it longer before adding solder.
- 16

Put solder on the pad labelled `3V3` .

☐

Third one down, right-hand edge.

You should see: a second shiny dome. **It must not touch the first one.**
- 17

Put solder on the pad labelled `D10` .

☐

Fourth down, right-hand edge.

You should see: three separate domes in a row.
- 18

Put solder on the pad labelled `D9` .

☐

Fifth down, right-hand edge.

You should see: four separate domes.
- 19

Put solder on the pad labelled `D8` .

☐

Sixth down, right-hand edge.

You should see: five separate domes.
- 20

Put solder on the pad labelled `D2` — on the OTHER edge.

☐

Left-hand edge, third one down. This is the only one on that side.

You should see: six domes total. Five on the right, one on the left.
- 21

Lower your eye to table level and look **ALONG the row of five.**

☐

Not down from above — from the side, like looking along a shelf.

You should see: five separate bumps with clear gaps between them.

If any two are joined: wipe the iron tip clean, then drag it firmly across the join and away off the edge of the board. The solder follows the tip. Look again from the side.

PART 4 — Attach the wires

The six real joints. Steps 22–29.

22 Put solder on ONE bare end of each of the six wires. ☐

Hold the iron under the twisted end, count two, touch solder to the wire. It soaks in. Do all six, then continue.

You should see: six wire ends gone stiff and silver instead of coppery and floppy.

If the plastic shrinks back: you held it too long. Still fine as long as bare metal isn't touching bare metal.

The joining move, for all six. Hold the silvered wire end flat against the silvered pad. Touch the iron on top of **both** for **one second** — the two blobs merge into one. Lift the iron. **Now do not move the wire for three seconds.**

23 Solder the BLACK wire to GND . ☐

You should see: one shiny joint. Tug the wire gently — it doesn't move.

If it looks dull, grey or crusty: you moved it while cooling. Just re-heat until it goes shiny, then hold still. Don't add more solder.

24 Solder the RED wire to 3V3 . ☐

You should see: two joints, not touching each other.

25 Solder the BLUE wire to D10 . ☐

You should see: three joints.

26 Solder the GREEN wire to D9 . ☐

You should see: four joints.

27 Solder the YELLOW wire to D8 . ☐

You should see: five joints, five wires leaving the right edge together.

28 Solder the WHITE wire to D2 on the left edge. ☐

You should see: six joints. That's every joint on the board done.

29 Tug each of the six wires gently, then look along the edge again. ☐

A firm pull, not a yank.

You should see: nothing moves, and no two joints are touching.

If a wire comes off: that joint was cold. Re-tin the pad and the wire end, and redo it.

PART 5 — Speaker, touch pad, amplifier

Last three joints. Steps 30–35.

- 30

Solder a short wire to one of the speaker's two metal tabs.
Any spare wire, about 4 cm. Same move: tin the tab, tin the wire, melt together.
You should see: one wire attached to the speaker.
Speaker already has wires? Skip steps 30 and 31 entirely.

☐
- 31

Solder a second short wire to the speaker's other tab.
It does not matter which wire goes to which tab. A single speaker has no correct way round.
You should see: two wires attached to the speaker.

☐
- 32

Rub sandpaper or a file over one face of your coin.
Any coin, or a wide screw head. Scratch it until it looks dull and scuffed, not shiny.
You should see: a dull, scratched patch about the size of a pea.
Why: solder will not stick to a shiny coin. This step is the whole reason the touch pad works.

☐
- 33

Solder the WHITE wire's free end to the scuffed patch.
A coin is a big lump of metal, so it takes longer to heat — 5–10 seconds, not 2. Be patient.
You should see: the wire stuck firmly to the coin. **That's all 9 joints done.**
If solder won't stick: not scuffed enough, or not hot enough. Scuff harder, hold the iron longer.

☐
- 34

Push the five long wires onto the amplifier's pins.
No soldering if your amp has pins sticking out — use jumper wires with plastic sockets, or just twist the bare wire tightly round each pin for now.
Red→ VIN · Black→ GND · Yellow→ BCLK · Green→ LRC · Blue→ DIN
You should see: five wires connected, each to the pin with the matching name.
Different names on your board? BCLK = SCK = CLK · LRC = WS = LRCLK · DIN = DATA .
Same signals.

☐
- 35

Connect the speaker's two wires to the amp's two output pins.
Labelled OUT+ and OUT−, or just + and −. Either wire to either pin.
You should see: the speaker connected to the amp.
Leave GAIN, SD and SHUTDOWN empty. Connecting SD silences the amp completely.

☐

PART 6 — Check it, then power it

Steps 36–42. Do not skip.

Everything is still unplugged. Steps 36–40 happen with no power anywhere.

- 36

Set your multimeter to the beep/continuity setting.

☐

The dial position marked with a sound-wave symbol ")))" or a diode triangle. Touch the two probes together — it should beep.

You should see: it beeps when the probes touch, stops when you part them.

No multimeter? Skip to step 41, but understand you're taking a real risk of killing the board.
- 37

Touch one probe to 3V3 and the other to GND .

☐

You should see: SILENCE. No beep.

If it beeps: STOP. Those two are joined by stray solder — a dead short. Find it (look along the edge), drag it off with a clean hot tip, and test again. **Do not plug in until this is silent.**
- 38

Touch one probe to D8 and the other to D9 .

☐

You should see: silence.

If it beeps: those two joints are bridged. Same fix as step 37.
- 39

Test D9 - D10 , then D10 - 3V3 .

☐

You should see: silence both times.
- 40

Now check each wire actually works: one probe on a pad, the other on that same wire's far end.

☐

Do all six. This proves the joint conducts, not just that it looks attached.

You should see: a beep every time. **This is the one test where beeping is good.**

If one doesn't beep: that joint looks fine but isn't connected. Re-heat it until the solder flows.
- 41

Plug the USB cable in.

☐

You should see: a small light on the board come on.

If nothing happens: unplug. Try the cable on something else — many USB cables are charge-only and carry no data.
- 42

Within 5 seconds, touch the board and the amp with a fingertip.

☐

Quick dab, don't hold your finger there.

You should see: both are cool or barely warm.

If anything is hot: UNPLUG IMMEDIATELY. You have a short. Go back to step 37 and test everything again.

Done — all 42 steps. The board is wired. Next: flash the firmware, start the server, and test the camera *before* the audio. Never both at once, or you won't know which one is broken.